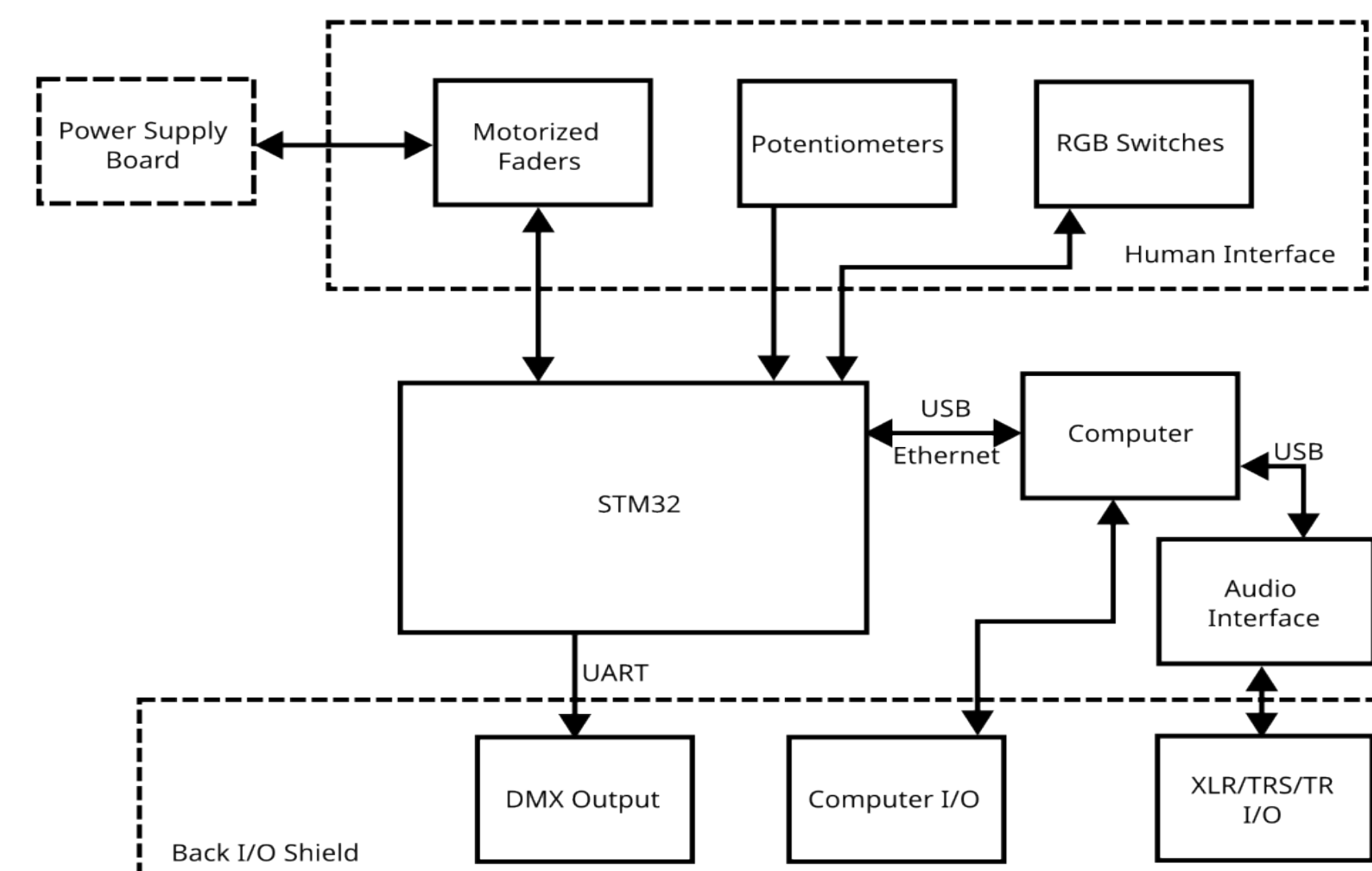


A/V Control Board

Brigham Young University – Idaho | ECEN 361 | Fall 2025

Aidan Pratt | Matthew Spotten | Jared Wightman | Ryan Christensen

Hardware Block Diagram



Project Plan

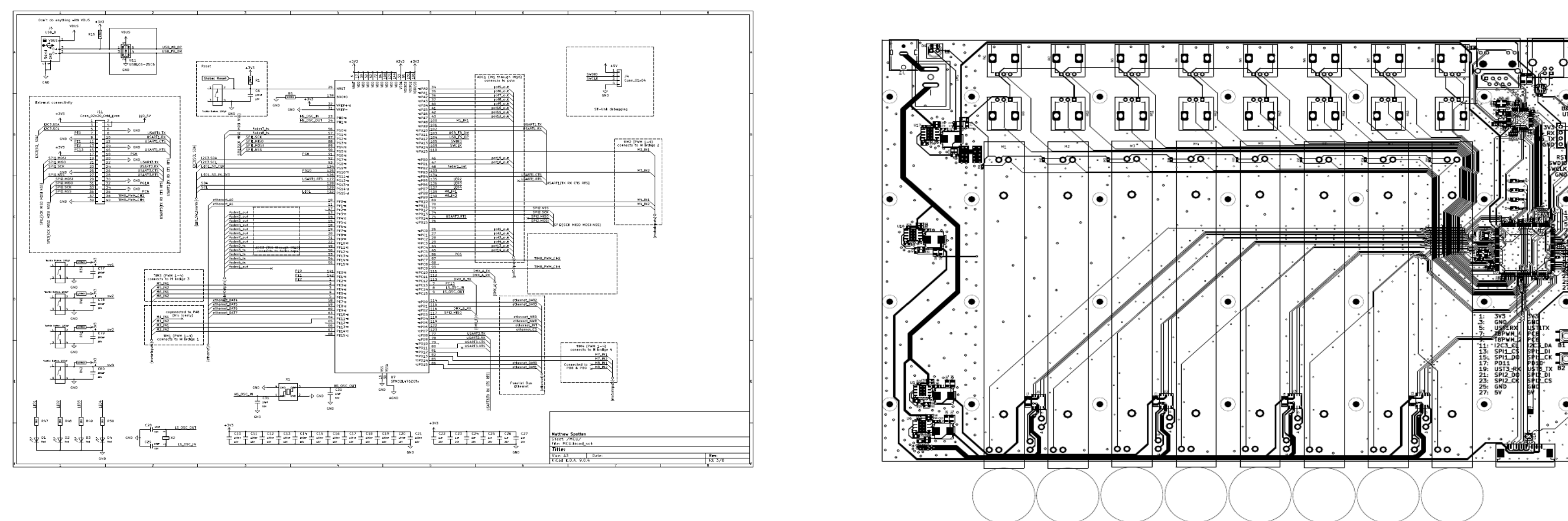
Our goal is to create a control board to compete in the audio/visual (AV) industry with. Current control boards for lighting and audio are typically too expensive (very few features for \$2,000, fully featured consoles for \$80,000 dollars). Our console is designed to be low-cost, compatible with a variety of input and output protocols, and compatible with existing proprietary and open source software.

Features

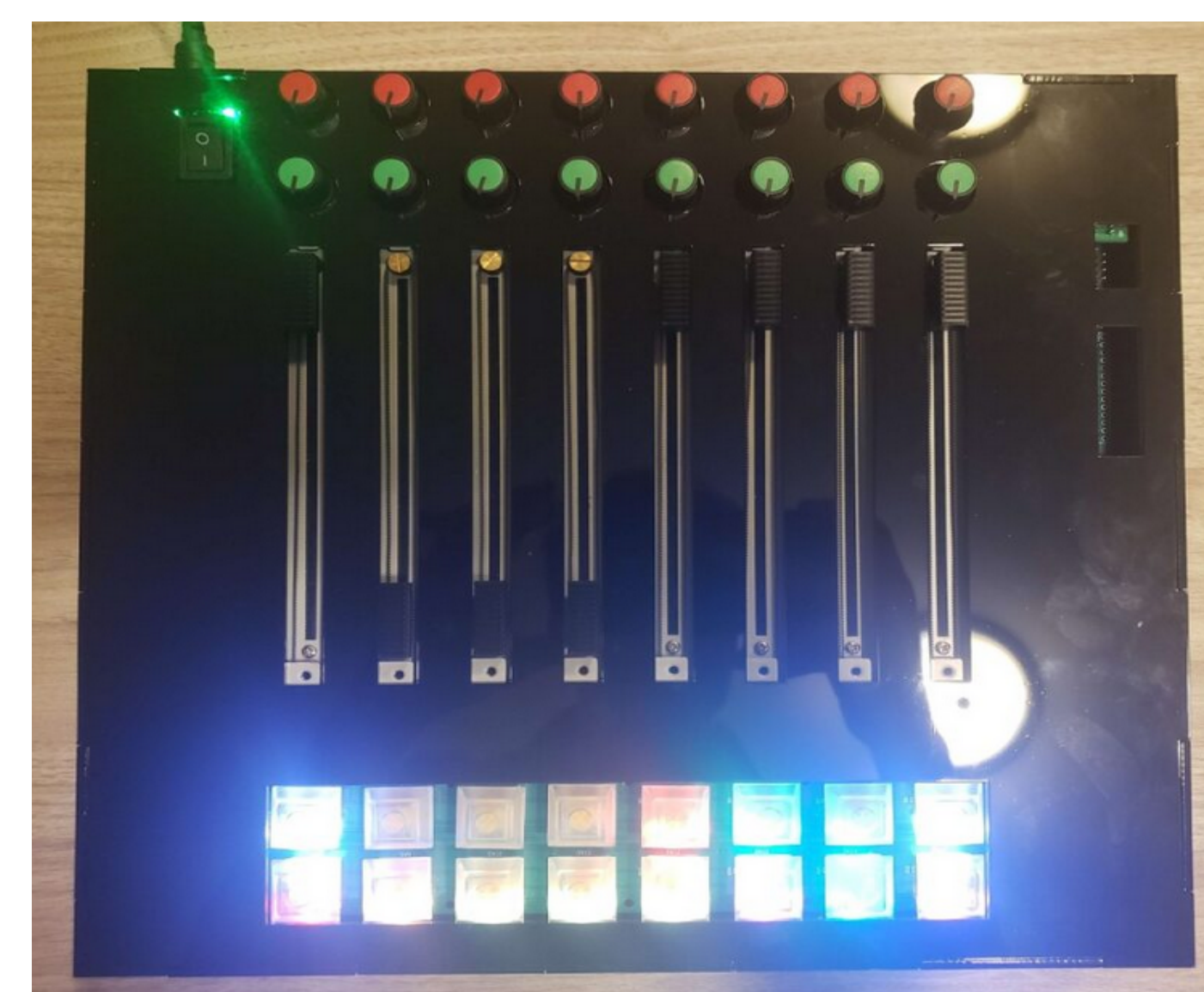
The board currently features:

- Ethernet
- USB
- 16 Potentiometers
- 8 Motorized Faders
- 16 RGB Key switches
- 2 DMX Outputs
- 2 DMX Inputs
- GPIO for expansion
- On-board LEDs (4) and buttons (2)

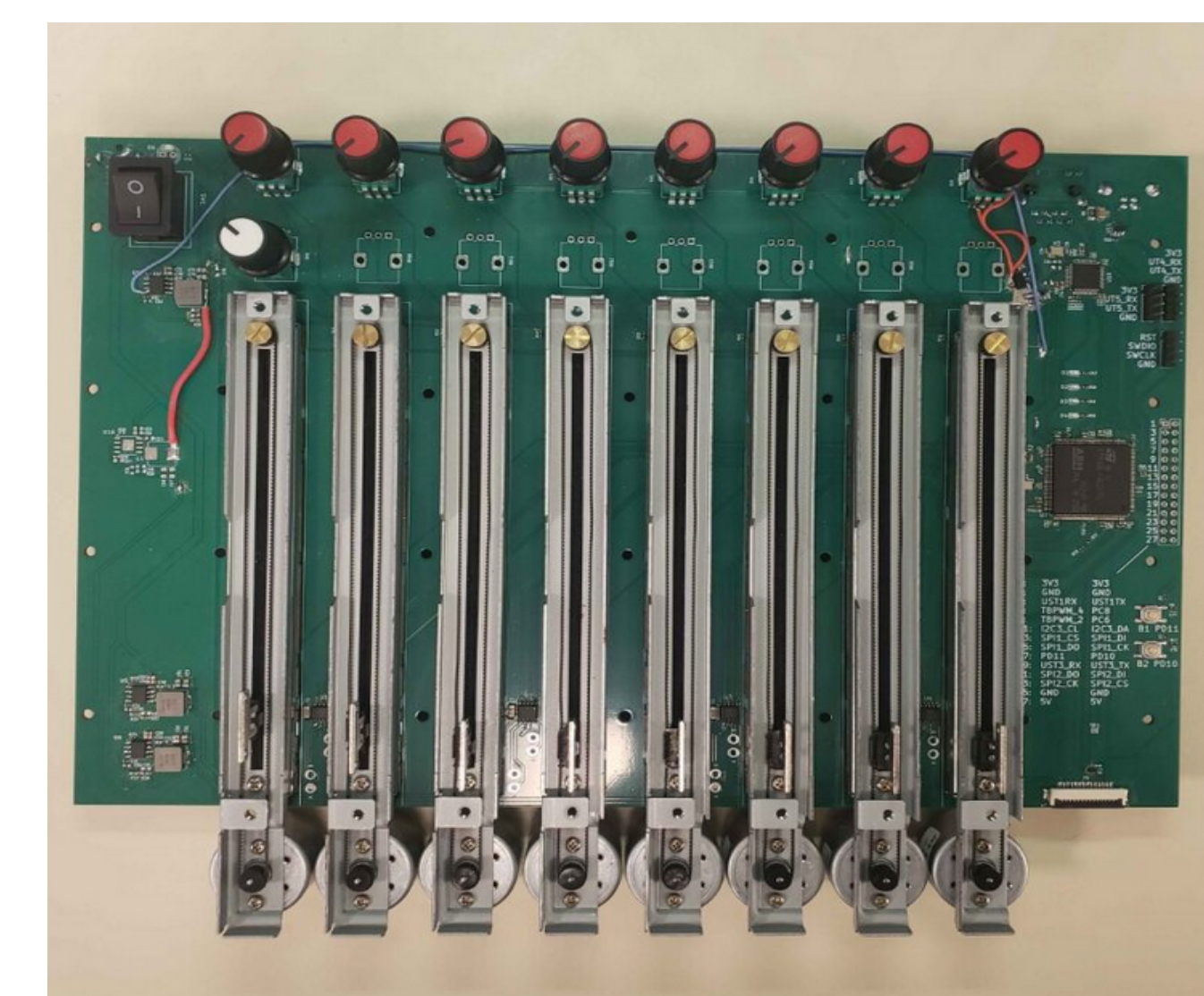
Kicad Drawings



Gallery



The final board in a case, powered on



The final board assembled

Future Steps

We plan to write a control application to assist with exposing different protocols from the board and to implement more protocols. We also plan to create a sheet metal enclosure and make a large console.

Manufacturing Cost

This is the price per board (it would likely be much lower with a higher quantity of boards).

Item	Price per board
Potentiometer caps	\$3.60
XLR Female connectors	\$7.91
Translucent Linear Switches	\$9.56
Potentiometers	\$15.98
Main board + components	\$26.59
Switch board + components	\$3.92
DMX board + components	\$1.50
Shipping JLCPCB	\$22.25
Shipping LCSC	\$4.73
Motorized Faders	\$8.40
Motorized Faders Shipping	\$8.64
Sum	\$113.08

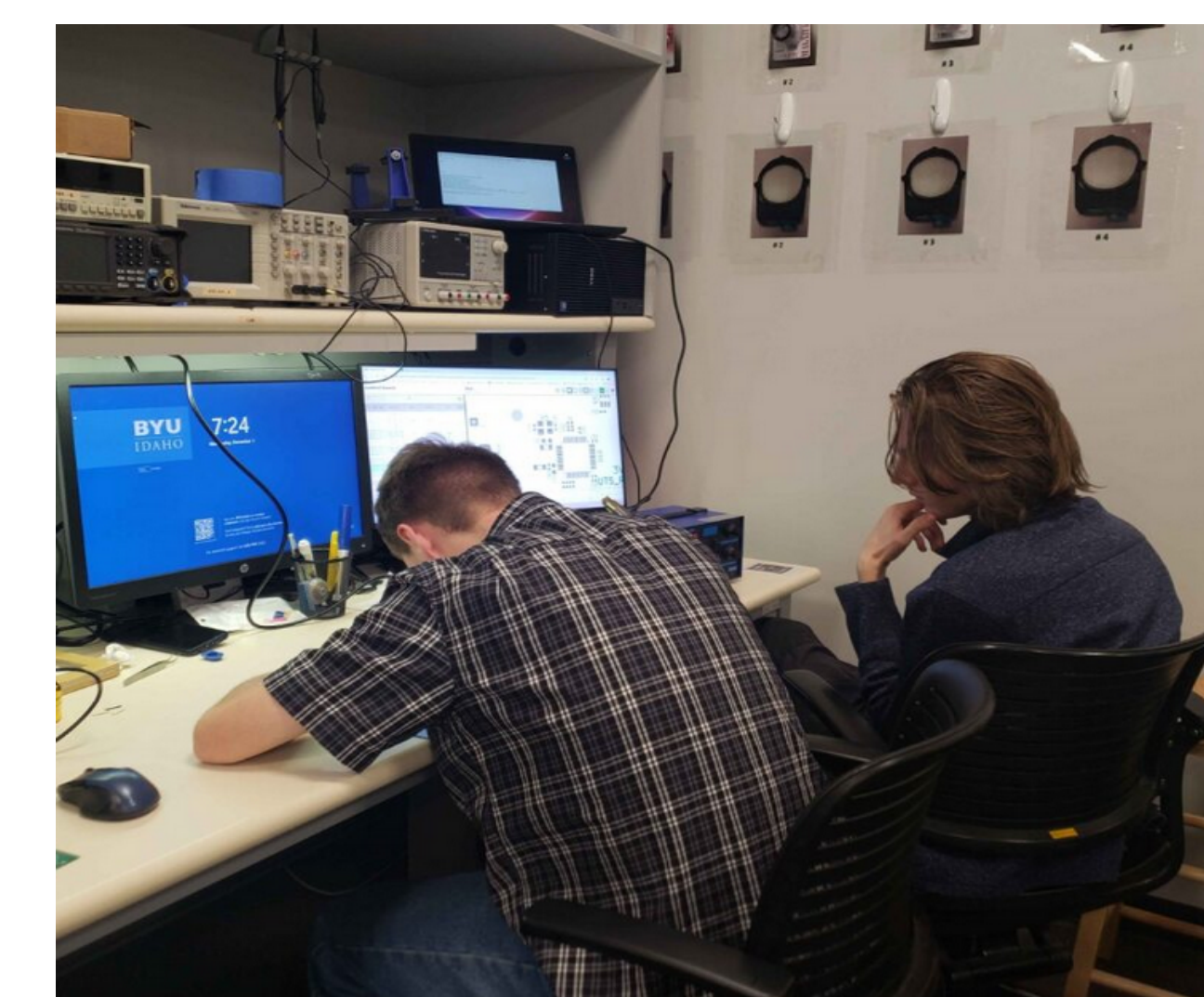
Communication

The board is designed to communicate with a host computer running audio or lighting control software over Ethernet or USB. It is capable of interfacing with multiple protocols simultaneously, such as MIDI, Open Sound Control (OSC), or ArtNet.

The board also has multiple DMX outputs and inputs available, making it so you can interface with stage lighting.



Aidan's sweatshop workers assembling the boards



Aidan hits the thinker pose while watching Matthew do work